

[Process Synchronization]

Considered there are N philosophers seated around a circular table with one chopstick between each pair of philosophers. There is one chopstick between each philosopher. A philosopher may eat if he can pick up the two chopsticks adjacent to him. One chopstick may be picked up by any one of its adjacent followers but not both.

Write a program to solve the problem using process synchronization technique.

Code:

```
/C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003
GNU nano 8.7 p6.c
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define N 5

sem_t chopstick[N];

void* philosopher(void* num) {
    int id = (int)num;

    printf("Philosopher %d is thinking\n", id);
    sleep(1);

    if (id == N - 1) {
        sem_wait(&chopstick[(id + 1) % N]);
        sem_wait(&chopstick[id]);
    } else {
        sem_wait(&chopstick[id]);
        sem_wait(&chopstick[(id + 1) % N]);
    }

    printf("Philosopher %d is eating\n", id);
    sleep(2);

    sem_post(&chopstick[id]);
    sem_post(&chopstick[(id + 1) % N]);

    printf("Philosopher %d finished eating\n", id);
    return NULL;
}

int main() {
    pthread_t phil[N];
    int ids[N];

    for (int i = 0; i < N; i++)
        sem_init(&chopstick[i], 0, 1);

    for (int i = 0; i < N; i++) {
        ids[i] = i;
        pthread_create(&phil[i], NULL, philosopher, &ids[i]);
    }
}
```

Output;

 /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003

```
Chanchal@DESKTOP-GARLJAI MSYS ~
# ^[[200~# cd /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003
-bash: $'\E[200~#': command not found

Chanchal@DESKTOP-GARLJAI MSYS ~
# cd /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003

Chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003
# nano p6.c

Chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003
# gcc p6.c -o p6
p6.c: In function 'philosopher':
p6.c:11:14: warning: cast from pointer to integer of different size [-Wpointer-to-int-cast]
   11 |         int id = (int)num;
      |                   ^

Chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003
# ./p6
Philosopher -13328 is thinking
Philosopher -13324 is thinking
Philosopher -13320 is thinking
Philosopher -13316 is thinking
Philosopher -13312 is thinking
Philosopher -13316 is eating
Philosopher -13312 is eating
Philosopher -13320 is eating
Philosopher -13328 is eating
Philosopher -13324 is eating
Philosopher -13320 finished eating
Philosopher -13316 finished eating
Philosopher -13324 finished eating
Philosopher -13328 finished eating
Philosopher -13312 finished eating

Chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003
# nano p6.c

Chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003
# |
```